

Module 1

Foundation Model Selection and Configuration

- **Core AWS resources**

Multi-LLM routing strategies for generative AI applications on AWS (Bedrock Intelligent Prompt Routing vs custom routing; includes reference architectures and patterns).

<https://aws.amazon.com/blogs/machine-learning/multi-llm-routing-strategies-for-generative-ai-applications-on-aws/>

- **Intelligent Prompt Routing in Amazon Bedrock – (serverless intelligent routing layer, Bedrock-native, dynamic model selection within a family).**

<https://builder.aws.com/content/2tSISEF1erMYLgFwB7fUNi9Gsbs/aws-serverless-intelligent-prompt-routing-in-amazon-bedrock-part-1-u>

- **Routing dynamic dispatch patterns – AWS Prescriptive Guidance (agentic LLM routing / dynamic dispatch reference patterns).**

<https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-patterns/routing-dynamic-dispatch-patterns.html>

- **Workflow for routing – AWS Prescriptive Guidance (workflow-level pattern for intent classification → route to model/agent).**

<https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-patterns/workflow-for-routing.html>

Module 2

Advanced Data Processing for Foundation Models

- **Amazon Bedrock Resources**

- **Amazon Bedrock User**

Guide: <https://docs.aws.amazon.com/bedrock/latest/userguide/>

- **Bedrock API**
Reference: <https://docs.aws.amazon.com/bedrock/latest/APIReference/>
- **Bedrock Model Parameters**
Guide: <https://docs.aws.amazon.com/bedrock/latest/userguide/model-parameters.html>
- **Bedrock Data**
Automation: <https://docs.aws.amazon.com/bedrock/latest/userguide/data-automation.html>
- **Bedrock Multimodal**
Capabilities: <https://docs.aws.amazon.com/bedrock/latest/userguide/multimodal-models.html>
- **Data Quality & Validation**
 - **AWS Glue Data Quality**: <https://docs.aws.amazon.com/glue/latest/dg/glue-data-quality.html>
 - **Glue Data Quality Rules**
Reference: <https://docs.aws.amazon.com/glue/latest/dg/dq-rules-reference.html>
 - **Amazon Comprehend Developer**
Guide: <https://docs.aws.amazon.com/comprehend/latest/dg/>
 - **Comprehend PII**
Detection: <https://docs.aws.amazon.com/comprehend/latest/dg/how-pii.html>
 - **Comprehend Custom**
Classification: <https://docs.aws.amazon.com/comprehend/latest/dg/how-document-classification.html>
- **Multi-Modal Processing Services**
 - **Amazon Transcribe Developer**
Guide: <https://docs.aws.amazon.com/transcribe/latest/dg/>
 - **Transcribe Real-time**
Streaming: <https://docs.aws.amazon.com/transcribe/latest/dg/streaming.html>

- **Amazon Textract Developer Guide:** <https://docs.aws.amazon.com/textract/latest/dg/>
- **Textract Document Analysis:** <https://docs.aws.amazon.com/textract/latest/dg/how-it-works-analyzing.html>

Hands-On Workshops & Labs

- **Amazon Bedrock Workshops**
 - **Bedrock Workshop:** <https://catalog.workshops.aws/amazon-bedrock/>
 - **RAG with Bedrock:** <https://catalog.workshops.aws/building-with-amazon-bedrock/>
- **Data Quality & Processing Workshops**
 - **Serverless Data Processing:** <https://catalog.workshops.aws/serverless-data-processing/>

Module 3

Vector Databases and Retrieval Augmentation

Encryption and Data Protection

- **Knowledge Base Encryption:** <https://docs.aws.amazon.com/bedrock/latest/userguide/encryption-kb.html>
- **Security Reference Architecture for RAG:** <https://docs.aws.amazon.com/prescriptive-guidance/latest/security-reference-architecture/gen-ai-rag.html>
- **AWS Prescriptive Guidance**
 - **Vector Database Selection Guide:** <https://docs.aws.amazon.com/prescriptive-guidance/latest/choosing-an-aws-vector-database-for-rag-use-cases/vector-db-options.html>
 - **RAG Implementation Options:** <https://docs.aws.amazon.com/prescriptive-guidance/latest/retrieval-augmented-generation-options/rag-fully-managed-bedrock.html>

- **Hands-On Learning Resources**
 - Multi-tenant RAG Implementation: <https://aws.amazon.com/blogs/machine-learning/multi-tenant-rag-implementation-with-amazon-bedrock-and-amazon-opensearch-service-for-saas-using-jwt/>
 - Vector Data Stores Deep Dive: <https://aws.amazon.com/blogs/machine-learning/dive-deep-into-vector-data-stores-using-amazon-bedrock-knowledge-bases/>
 - OpenSearch as Vector Database: <https://aws.amazon.com/blogs/big-data/improve-search-results-for-ai-using-amazon-opensearch-service-as-a-vector-database-with-amazon-bedrock/>
- **S3 Vectors Integration**
 - S3 Vectors with Bedrock: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-bedrock-kb.html>
 - S3 Vectors Overview: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors.html>

Module 4

Prompt Engineering and Governance

Prompt Engineering & Management

- **Prompt Management in Bedrock:** <https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html>
- **Prompt Engineering Guide:** <https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-engineering-guidelines.html>
- **Bedrock Flows:** <https://docs.aws.amazon.com/bedrock/latest/userguide/flows.html>

- **Structured Data Response Blog:** <https://aws.amazon.com/blogs/machine-learning/structured-data-response-with-amazon-bedrock-prompt-engineering-and-tool-use/>

Chain-of-Thought & Complex Reasoning

- **Structured Outputs with Amazon Nova:** <https://aws.amazon.com/blogs/machine-learning/structured-outputs-with-amazon-nova-a-guide-for-builders/>
- **Advanced Prompt Techniques:** <https://docs.aws.amazon.com/bedrock/latest/userguide/advanced-prompts.html>
- **Conversational AI Patterns:** <https://aws.amazon.com/blogs/machine-learning/build-a-contextual-chatbot-application-using-knowledge-bases-for-amazon-bedrock/>

Enterprise Governance & Security

- **IAM and Access Control**
 - **AWS IAM User Guide:** <https://docs.aws.amazon.com/IAM/latest/UserGuide/>
 - **Bedrock IAM Permissions:** <https://docs.aws.amazon.com/bedrock/latest/userguide/security-iam.html>
 - **IAM Best Practices:** <https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html>
- **Monitoring & Logging**
 - **Amazon CloudWatch User Guide:** <https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/>
 - **AWS CloudTrail User Guide:** <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/>
 - **Bedrock Monitoring:** <https://docs.aws.amazon.com/bedrock/latest/userguide/monitoring.html>

- **CloudWatch Metrics for Bedrock:** <https://docs.aws.amazon.com/bedrock/latest/userguide/cloudwatch-metrics.html>
- **Compliance & Audit**
 - **AWS Artifact:** <https://aws.amazon.com/artifact/>
 - **AWS Compliance Center:** <https://aws.amazon.com/compliance/>
 - **Well-Architected Framework - AI/ML Lens:** <https://docs.aws.amazon.com/wellarchitected/latest/machine-learning-lens/>

Module 5

Implementing Agentic AI Frameworks with Amazon Bedrock AgentCore

Amazon Bedrock AgentCore Resources

- **Core Documentation:**
 - Amazon Bedrock AgentCore Developer Guide: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/what-is-bedrock-agentcore.html>
 - Getting Started with AgentCore: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/agentcore-get-started-toolkit.html>
 - AgentCore Starter Toolkit: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/runtime-get-started-toolkit.html>
- **AgentCore Services Documentation:**
 - AgentCore Runtime: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/agents-tools-runtime.html>
 - AgentCore Gateway: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway.html>
 - AgentCore Observability: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/observability.html>

- AgentCore Memory Examples: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/memory-examples.html>
- **Framework Integration:**
 - Using Any Agent Framework with AgentCore: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/using-any-agent-framework.html>
 - AgentCore MCP Server: <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/mcp-getting-started.html>

AWS Prescriptive Guidance for Agentic AI

- **Framework Comparisons:**
 - Agentic AI Frameworks Guide: <https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-frameworks/introduction.html>
 - Framework Comparison: <https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-frameworks/frameworks.html>
 - Strands Agents Documentation: <https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-frameworks/strands-agents.html>
- **Deployment Patterns:**
 - CrewAI with Terraform: <https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/deploy-agentic-systems-on-amazon-bedrock-with-the-crewai-framework.html>
 - Agentic Workflow Design: <https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-patterns/designing-agentic-workflows-on-aws.html>

Amazon Bedrock Agents Resources

- **Core Bedrock Agents:**
 - Amazon Bedrock Agents Documentation: <https://aws.amazon.com/bedrock/agents/>
 - Bedrock Agents API Reference: https://docs.aws.amazon.com/bedrock/latest/APIReference/API_agent_Agent.html

- Bedrock Agents Code Examples: https://docs.aws.amazon.com/code-library/latest/ug/bedrock-agent_code_examples_basics.html

AWS Blog Resources

- **Technical Deep Dives:**

- AgentCore Runtime Blog: <https://aws.amazon.com/blogs/machine-learning/securely-launch-and-scale-your-agents-and-tools-on-amazon-bedrock-agentcore-runtime/>
- AgentCore Observability Blog: <https://aws.amazon.com/blogs/machine-learning/build-trustworthy-ai-agents-with-amazon-bedrock-agentcore-observability/>
- AgentCore Code Interpreter: <https://aws.amazon.com/blogs/machine-learning/introducing-the-amazon-bedrock-agentcore-code-interpreter/>
- Multi-Agent Systems: <https://aws.amazon.com/blogs/machine-learning/build-multi-agent-site-reliability-engineering-assistants-with-amazon-bedrock-agentcore/>

- **Framework-Specific Blogs:**

- CrewAI with Bedrock: <https://aws.amazon.com/blogs/machine-learning/build-agentic-systems-with-crewai-and-amazon-bedrock/>
- Strands Agents Deep Dive: <https://aws.amazon.com/blogs/machine-learning/amazon-strands-agents-sdk-a-technical-deep-dive-into-agent-architectures-and-observability/>

- **Hands-On Examples:**

- Healthcare Agents: <https://aws.amazon.com/blogs/machine-learning/building-health-care-agents-using-amazon-bedrock-agentcore/>
- Research Agents: <https://aws.amazon.com/blogs/machine-learning/running-deep-research-ai-agents-on-amazon-bedrock-agentcore/>

Module 6

AI Safety and Security

- **Core Documentation**
 - **Amazon Bedrock Guardrails User Guide**
<https://docs.aws.amazon.com/bedrock/latest/userguide/guardrails.html>
 - **Amazon Bedrock Security Best Practices**
<https://docs.aws.amazon.com/bedrock/latest/userguide/security-best-practices.html>
 - **Bedrock Guardrails API Reference**
https://docs.aws.amazon.com/bedrock/latest/APIReference/API_Operations_Amazon_Bedrock_Runtime.html
- **Hands-On Tutorials**
 - **Getting Started with Amazon Bedrock Guardrails**
<https://aws.amazon.com/blogs/machine-learning/getting-started-with-amazon-bedrock-guardrails/>
 - **Building Responsible AI with Amazon Bedrock Guardrails**
<https://aws.amazon.com/blogs/machine-learning/building-responsible-ai-with-amazon-bedrock-guardrails/>

Security & Threat Detection

- **AWS Security Hub User Guide**
<https://docs.aws.amazon.com/securityhub/latest/userguide/what-is-securityhub.html>
- **Amazon GuardDuty User Guide**
<https://docs.aws.amazon.com/guardduty/latest/ug/what-is-guardduty.html>
- **AWS WAF Developer Guide**
<https://docs.aws.amazon.com/waf/latest/developerguide/waf-chapter.html>
- **PII Detection & Data Protection**
 - **Amazon Comprehend PII Detection**
<https://docs.aws.amazon.com/comprehend/latest/dg/how-pii.html>
 - **Amazon Macie User Guide**
<https://docs.aws.amazon.com/macie/latest/user/what-is-macie.html>

- **Data Protection in Amazon Bedrock**
<https://docs.aws.amazon.com/bedrock/latest/userguide/data-protection.html>
- **Access Controls & Zero Trust**
 - **IAM Identity Center User Guide**
<https://docs.aws.amazon.com/singlesignon/latest/userguide/what-is.html>
 - **AWS Zero Trust Architecture Whitepaper**
<https://docs.aws.amazon.com/whitepapers/latest/zero-trust-architectures/zero-trust-architectures.html>
 - **VPC Endpoints User Guide**
<https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints.html>

AI GOVERNANCE & COMPLIANCE

- **Model Documentation & Transparency**
 - **Amazon SageMaker Model Cards**
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-cards.html>
 - **SageMaker Model Registry**
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-registry.html>
 - **Bias Detection & Explainability**
 - **Amazon SageMaker Clarify**
<https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-fairness-and-explainability.html>
 - **Bias Detection and Model Explainability**
<https://aws.amazon.com/blogs/machine-learning/bias-detection-and-model-explainability-amazon-sagemaker-clarify>
- **Human-in-the-Loop & Feedback Systems**
 - **Amazon Augmented AI (A2I)**
<https://docs.aws.amazon.com/sagemaker/latest/dg/a2i-use-augmented-ai-a2i-human-review-loops.html>
 - **A2I Getting Started Tutorial**
<https://aws.amazon.com/blogs/machine-learning/getting-started-with-amazon-augmented-ai-a2i/>

COMPLIANCE & REGULATORY FRAMEWORKS

- **Regulatory Compliance**
 - **AWS GDPR Center**
<https://aws.amazon.com/compliance/gdpr-center/>
 - **AWS HIPAA Compliance**
<https://aws.amazon.com/compliance/hipaa-compliance/>
 - **AWS SOC Compliance**
<https://aws.amazon.com/compliance/soc/>
- **AI Ethics & Responsible AI**
 - **AWS Responsible AI Practices**
<https://aws.amazon.com/machine-learning/responsible-ai/>
 - **AI Ethics Whitepaper**
<https://docs.aws.amazon.com/whitepapers/latest/model-explainability-aws-ai-ml/model-explainability-aws-ai-ml.html>

HANDS-ON LABS & WORKSHOPS

- **Interactive Learning**
 - **Amazon Bedrock Workshop**
<https://catalog.us-east-1.prod.workshops.aws/workshops/a4bdb007-5600-4368-81c5-ff5b4154f518>
- **Code Examples & Samples**
 - **Amazon Bedrock Samples Repository**
<https://github.com/aws-samples/amazon-bedrock-samples>
 - **SageMaker Examples Repository**
<https://github.com/aws/amazon-sagemaker-examples>

Module 7

Performance Optimization and Cost Management

Core AWS Documentation

- **Model Selection**
Guide: <https://docs.aws.amazon.com/bedrock/latest/userguide/model-parameters.html>

- Token Usage and Optimization: <https://docs.aws.amazon.com/bedrock/latest/userguide/model-parameters-claude.html>

Cost Management and Monitoring

- AWS Cost Management: <https://docs.aws.amazon.com/cost-management/latest/userguide/what-is-costmanagement.html>
- AWS Budgets: <https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html>
- Cost Anomaly Detection: <https://docs.aws.amazon.com/cost-management/latest/userguide/getting-started-ad.html>
- AWS Cost Explorer: <https://docs.aws.amazon.com/cost-management/latest/userguide/ce-what-is.html>

Caching and Performance Resources

- Amazon ElastiCache for Semantic Caching
 - ElastiCache for Redis User Guide: <https://docs.aws.amazon.com/AmazonElastiCache/latest/red-ug/WhatIs.html>
 - Redis Vector Search: <https://docs.aws.amazon.com/AmazonElastiCache/latest/red-ug/vector-search.html>
 - MemoryDB for Redis: <https://docs.aws.amazon.com/memorydb/latest/devguide/what-is-memorydb-for-redis.html>
 - Caching Best Practices: <https://docs.aws.amazon.com/AmazonElastiCache/latest/red-ug/BestPractices.html>
- Amazon CloudFront for Global Distribution
 - CloudFront Developer Guide: <https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Introduction.html>

- **Cache Behaviors**
Configuration: <https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/distribution-web-values-specify.html#DownloadDistValuesCacheBehavior>
- **CloudFront Pricing:** <https://aws.amazon.com/cloudfront/pricing/>
- **Auto-Scaling and Architecture Resources**
 - **Amazon SageMaker Auto-Scaling**
 - **SageMaker Auto**
Scaling: <https://docs.aws.amazon.com/sagemaker/latest/dg/endpoint-auto-scaling.html>
 - **Multi-Model**
Endpoints: <https://docs.aws.amazon.com/sagemaker/latest/dg/multi-model-endpoints.html>
 - **SageMaker Inference**
Optimization: <https://docs.aws.amazon.com/sagemaker/latest/dg/inference-optimization.html>
 - **AWS Application Auto Scaling**
 - **Application Auto Scaling User**
Guide: <https://docs.aws.amazon.com/autoscaling/application/userguide/what-is-application-auto-scaling.html>
 - **Predictive**
Scaling: <https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-predictive-scaling.html>

Module 8

Monitoring and Observability for Generative AI

- **Amazon Bedrock**
 - **Main Documentation:** <https://docs.aws.amazon.com/bedrock/>
 - **Model Invocation Logging:**
<https://docs.aws.amazon.com/bedrock/latest/userguide/model-invocation-logging.html>

- **Monitoring and Logging:**
<https://docs.aws.amazon.com/bedrock/latest/userguide/monitoring.html>
- **API Reference:** <https://docs.aws.amazon.com/bedrock/latest/APIReference/>
- **Amazon CloudWatch**
 - **Main Documentation:** <https://docs.aws.amazon.com/cloudwatch/>
 - **Custom Metrics:**
<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/publishingMetrics.html>
 - **CloudWatch Dashboards:**
<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/CloudWatchDashboards.html>
 - **CloudWatch Alarms:**
<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/AlarmThatSendsEmail.html>
 - **CloudWatch Logs Insights:**
<https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/AnalyzingLogData.html>
 - **Anomaly Detection:**
<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/CloudWatchAnomalyDetection.html>
- **SageMaker Model Monitoring**
 - **Model Monitor:** <https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor.html>
 - **Data Quality Monitoring:**
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-data-quality.html>
 - **Model Quality Monitoring:**
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-model-quality.html>
- **AWS Well-Architected ML Lens**

- ML Lens Whitepaper: <https://docs.aws.amazon.com/wellarchitected/latest/machine-learning-lens/welcome.html>
- Operational Excellence for ML: <https://docs.aws.amazon.com/wellarchitected/latest/machine-learning-lens/operational-excellence.html>
- Observability Best Practices
 - AWS Observability Best Practices: <https://aws-observability.github.io/observability-best-practices/>
 - Application Monitoring: <https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/appinsights-what-is.html>
 - Distributed Tracing with X-Ray: <https://docs.aws.amazon.com/xray/latest/devguide/aws-xray.html>

Hands-On Learning Resources

- CloudWatch Workshops: <https://catalog.workshops.aws/observability>

Module 9

Testing, Validation, and Continuous Improvement

- Amazon SageMaker - Model Monitoring & Quality Gates
 - Amazon SageMaker Model Monitor: <https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor.html>
 - SageMaker Model Monitor - Data Quality: <https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-data-quality.html>
 - SageMaker Model Monitor - Model Quality: <https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-model-quality.html>

- SageMaker Model Monitor - Bias Drift: <https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-model-monitor-bias-drift.html>
- AWS Step Functions - Automated Evaluation Pipelines
 - Core Documentation
 - AWS Step Functions Developer Guide: <https://docs.aws.amazon.com/step-functions/latest/dg/welcome.html>
 - Step Functions Express Workflows: <https://docs.aws.amazon.com/step-functions/latest/dg/concepts-express-workflows.html>
 - Step Functions Error Handling: <https://docs.aws.amazon.com/step-functions/latest/dg/concepts-error-handling.html>

Core MLOps Documentation

- Amazon SageMaker MLOps: <https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-projects.html>
- SageMaker Pipelines: <https://docs.aws.amazon.com/sagemaker/latest/dg/pipelines.html>
- SageMaker Feature Store: <https://docs.aws.amazon.com/sagemaker/latest/dg/feature-store.html>
- AWS CodePipeline for ML: <https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-projects-walkthrough.html>
- AWS CodeCommit: <https://docs.aws.amazon.com/codecommit/latest/userguide/welcome.html>

Module 10

Enterprise Integration Patterns

Hands-On Workshops and Tutorials

- AWS Workshops

- API Gateway Workshop: <https://catalog.workshops.aws/apigateway/en-US>
- Serverless Workshop: <https://catalog.workshops.aws/serverless-patterns/en-US>
- AWS Samples and Code Examples
- Serverless Patterns: <https://serverlessland.com/patterns>
- AWS Samples GitHub: <https://github.com/aws-samples>

Security and Compliance Resources

- AWS WAF
 - Developer Guide: <https://docs.aws.amazon.com/waf/latest/developerguide/waf-chapter.html>
 - Rate-Based Rules: <https://docs.aws.amazon.com/waf/latest/developerguide/waf-rule-statement-type-rate-based.html>
- AWS Secrets Manager
 - User Guide: <https://docs.aws.amazon.com/secretsmanager/latest/userguide/intro.html>
 - Rotation: <https://docs.aws.amazon.com/secretsmanager/latest/userguide/rotating-secrets.html>